1 FIG-211 — Neb topology audit

Group: NEB-TOPOLOGY-AUDIT **Evidence chain:** manifested **Status:** historical/reconstruction audit; not an accepted DFT barrier figure

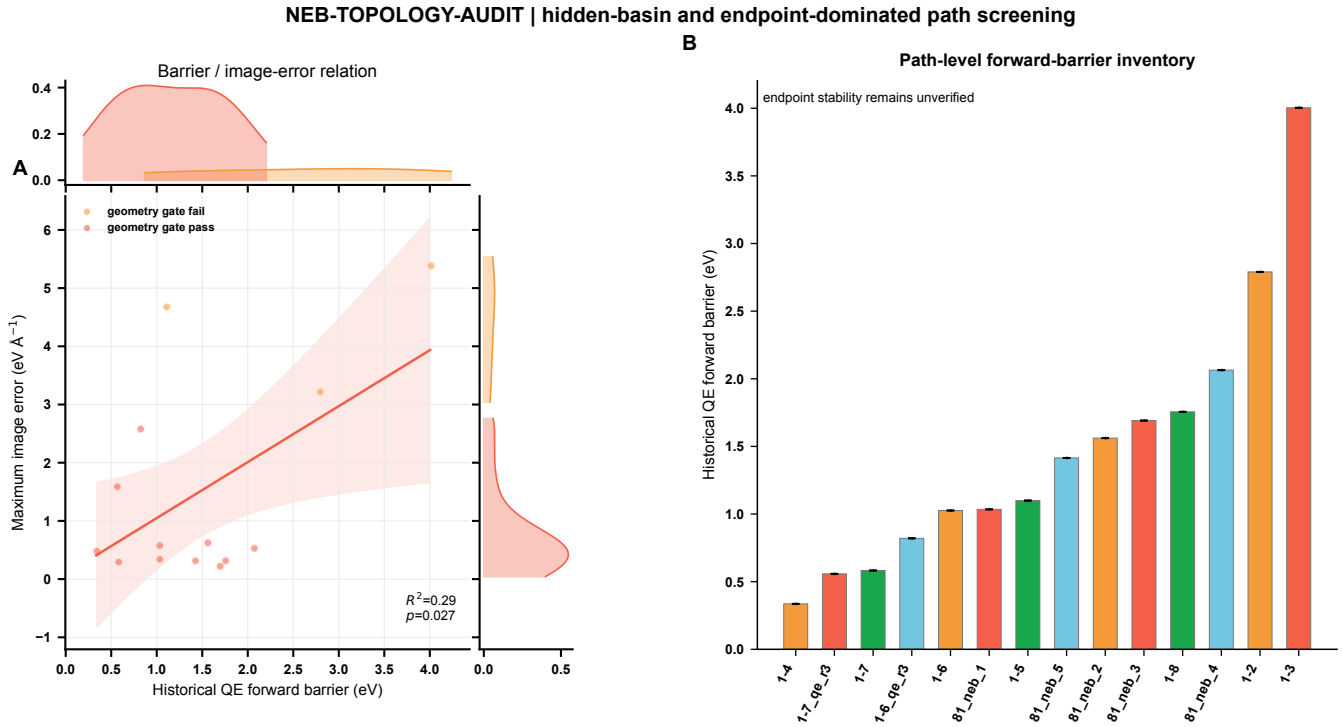


Figure 1: A, historical QE forward-barrier values versus maximum image force error, colored by the geometry-gate flag; B, sorted path-level forward-barrier inventory. This page shows the Plot Atlas table-backed candidate "Neb topology audit"; its scientific status and caveats are reproduced from the package manifest.

Results text draft. The table contains historical and 81-atom path entries with hidden-basin candidates and endpoint-dominated segments. Several paths have geometry-gate failure or large image errors, while the 0.336 eV 1-4 value is retained as a candidate path value rather than an accepted barrier. **Discussion and boundary.** This figure is a topology/provenance audit for deciding which path segments require independent endpoint relaxation. It does not certify endpoint stability, NEB convergence, or a publication barrier ranking. **Article placement.** Proposed article location: SI Detailed Migration Pathway Analysis: historical path topology and hidden-basin audit. **Reviewer linkage:** overall status PARTIAL.

- **REV-A1 (PARTIAL)**: Reconcile Fig. 3d fixed-geometry DFT-path scoring with SI Fig. S3 self-consistent MLFF-NEB, including pathway identity and the DFT reference. *Completed:* The fixed-path versus self-relaxed operator distinction and the narrower model-reported-barrier wording are integrated. *Still required:* Close the restartable QE path-1–4 convergence gate, preserve the final path/source metadata, and complete the SI Fig. S3 endpoint/checkpoint/optimizer/source audit before calling the quantitative reference final.
- **REV-A3 (PARTIAL)**: Add seed-level uncertainty and test whether the deep-penetration result is seed-stable rather than a chance outcome. *Completed:* Three-seed grouped in-gap comparisons and uncertainty are available for the declared Scratch/FT-600K/Scratch-5% scope. *Still required:* Complete path-resolved deep-penetration classification and restarted QE references before reporting a pooled deep-penetration ranking.

Evidence pointers.

- Data/processed: paper/data/published/path_topology_audit/path_topology_summary.csv
- Data/processed: paper/data/published/path_topology_audit/segmented_barriers.csv
- Plot/processing script: main/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py

- Caption/text source: main/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/NEB-TOPOLOGY-AUDIT/caption_results_discussion.md

Machine-collected evidence candidates (generated from the audit matrix; do not edit this file).

- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/NEB-TOPOLOGY-AUDIT/caption_results_discussion.md
- **Raw or processed data:** /Users/alina/Project/migrationbench-paper-repro/data/published/path_topology_audit/path_topology_summary.csv
- **Raw or processed data:** /Users/alina/Project/migrationbench-paper-repro/data/published/path_topology_audit/segmented_barriers.csv
- **Raw or processed data:** /Users/alina/Project/migrationbench-paper-repro/data/published/path_topology_audit/README.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/NEB-TOPOLOGY-AUDIT/processed_topology_summary.csv
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py
- **Plot or processing script:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: Endpoint stability is explicitly unverified because inspected endpoints are frozen. The geometry flag is not a convergence certificate. Keep this family separate from current Cr-61/Cr-81 exploratory path figures.